

DCN Message format

DCN messages use a human-readable, simple ASCII format. This allows messages to be easily sent and read using a terminal program, such as PuTTY or similar.

General Message format:

```
/FFT:CMD,P1,P2,P3<etc>:XX<cr>
```

Where:

- '/' is required as the first character of the message
- FF is the address of the device sending the message. This can be any 2 characters, but is typically a 2 digit number. By convention, the master controller has an address of 00.
- TT is the address of the device that the message is directed to. It is the same format as the 'from' address.
- ':' is required to separate the addresses from the main message content.
- CMD is the command or message type string. (See supported commands below)
- P1, P2, P3, etc. are the message parameters. These are specific to each command or message type. Parameters are separated by commas.
The maximum number of parameters supported varies by device.

NOTE: The Command and Parameters combined are the message Payload and fall between the 2 ':' characters.

- ':' is a required separator character.
- 'XX' is a future CRC error check. Currently, it is the literal characters 'XX'
- <cr> - Carriage Return terminates the command string.

Special broadcast message format:

```
//CMD,P1<etc><cr>
```

- '/' - 2 consecutive start characters indicate a broadcast message. No From or To addresses are included. All devices on the network will respond to this message.
- CMD,P1 – Command and parameters follow the same format as described above.
- <cr> - The message is terminated with a Carriage Return. There is no terminating separator nor CRC characters.

The broadcast message format can be used to send a command to all devices. It is also typically used for programming or configuring a device that is directly connected to a computer with no other devices connected.

SPS-1 DCN Commands

Summary of Commands

COMMAND	Parameters	Description
PWR	0 or 1	Power output Enable
STATE	None	Returns the current operating state in UPDATE message
UVSET	Millivolts	Sets the under-voltage trip level in millivolts
OVSET	Millivolts	Sets the over-voltage trip level in millivolts
OCSET	Milliamps, Reset type, Delay	Sets over-current trip level in milliamps and over-current behavior.
MOBSET	Enable, Off-Millivolts On-Millivolts Timeout min.	Sets Mobile Mode operation parameters.
SWMODE	0 or 1	Configures local enable switch to be in momentary mode or not.
CONFIG	None	Returns current configuration settings in SETTINGS message
VERSION	None	Returns current firmware version in FWVER message (V1.1)
CALLO	Milliamps	Calibration Low current setting IMPORTANT: Read calibration section before using this command.
CALHI	Milliamps	Calibration High current setting IMPORTANT: Read calibration section before using this command.
CALSET	Scale Setting Offset Setting	Updates the stored calibration setting to the values provided. Can be used to restore calibration parameters if EEPROM data is cleared or becomes corrupted. (V1.1)
LOGS	None	Returns SPS-1 log data
SETADDR	2 character DCN address	Sets device DCN address
RSTLOGS	None	Resets log data (except total on time)
REBOOT	None	Forces a processor restart – equivalent to removing and restoring power. (V1.1)

NOTE: The SPS-1 ignores invalid commands or any message that is not formatted correctly. There is no response or direct acknowledgement of operational commands. The execution of PWR and the various SET commands can be confirmed by checking the response to STATE, CONFIG, or LOGS commands.

Command Details

- **PWR,n**

n=1 requests output power ON

n=0 requests output power OFF

Power will only be turned ON if the input voltage is within Undervoltage & Overvoltage limits.

This command works in conjunction with the local, hardwired Enable signal. If either the local enable or the DCN Enable is on (1), then power will be enabled. BOTH enable signals must be OFF to turn power off.

- **STATE** <no parameters>

Returns the current state of the SPS1 in the following format:

/fftt:UPDATE,SPS1,R,P,FS,r,VV.VVV,AA.AAA,WD:XX<cr>

Where:

- ff = The address of this device
- tt = The address of the device that sent the STATE command (or 00 if State was sent in a broadcast message)
- UPDATE = Response message identifier
- SPS1 = The name of this device
- R = 1 Power is Requested ON, 0 if power is requested OFF (either Local or DCN Request)
- P = 1 if output power switch is ON, 0 if output power switch is OFF
- FS = Fault Status – “ ”- No Fault, “UV”- Undervoltage Fault, “OV”- Overvoltage Fault
“OC”- Overcurrent Fault
- r = The number of OverCurrent reset retries that have been attempted.
(NOTE: This only applies if OverCurrent Auto Reset is set to TRUE, otherwise this value will be zero)
- VV.VVV – The voltage on the power input terminals
- AA.AAA – The current in amps flowing through the switch
- WatchDog status -

This field should normally be “--”. If this field contains WD, then the internal controller has experienced a WatchDog reset. The FAULT LED will flash rapidly and the Output will be turned off if this occurs. The DCN and Local Enable signals must be turned Off to reset the switch. Then the switch will operate normally again. The FAULT LED flashing will be reset the next time the Output is turned on.

“WD” will remain in this field until power is removed from the SPS-1 or until a REBOOT command is executed.

NOTE: Please notify A2Z Tech if this condition occurs.

- **UVSET,nnnnn**

Sets Undervoltage trip limit

nnnnn= trip limit in millivolts.

This command will ONLY be processed if the switch is in the IDLE state (enables OFF). Otherwise it is ignored.

Value must be between 6000 and 18000. Any other values or non-numeric characters are invalid and the command will be ignored.

The result of this command is written to EEPROM and persists through power cycles. The new undervoltage limit takes effect immediately.

- **OVSET,nnnnn**

Sets Overvoltage trip limit

nnnnn= trip limit in millivolts.

This command will ONLY be processed if the switch is in the IDLE state (enables OFF). Otherwise it is ignored.

Value must be between 6000 and 18000. Any other values or non-numeric characters are invalid and the command will be ignored.

The result of this command is written to EEPROM and persists through power cycles. The new overvoltage limit takes effect immediately.

- **OCSET,nnnnn,a,t**

Sets Overcurrent trip limit

nnnnn= trip limit in milliamps

NOTE: The minimum trip current allowed is 5000 and the maximum allowed is 35000.

a= Automatic Reset selection – 0= Manual Reset 1= Automatic Reset

t= Reset Delay in seconds (Whole seconds only. No decimals)

When configured for Manual Reset an overcurrent trip is reset by toggling the enable OFF and then back ON.

This command will ONLY be processed if the switch is in the IDLE state (enables OFF). Otherwise it is ignored.

The result of this command is written to EEPROM.

IMPORTANT: The new over-current parameters take effect immediately. However, unless the SET switch is set to the 0 position, the over-current level will be reset based on SET switch position whenever input power to the SPS-1 is removed and restored.

- **MOBSET,e,mmmmm,nnnnn,ttt**
Mobile Mode Settings

e= 0 to Disable Mobile Mode, 1 to Enable Mobile Mode

mmmmm= Off Threshold in mv. If voltage dips below this voltage, the timeout timer starts counting down.

NOTE: This must be at least 1V (1000mv) greater than the Under Voltage trip level.

nnnnn= On Threshold in mv. When the voltage goes above this level, the output will be turned back on if the timeout has expired.

If the output is on and timeout timer is counting down, the timer will be cancelled.

NOTE: This must be at least 1V (1000mv) less than the Over Voltage trip level.

ttt= Timeout Timer period in minutes. This is how long the output will remain on once input voltage dips below the Off Threshold. This value must be at least 1 and no more than 100 minutes.

This command will ONLY be processed if the switch is in the IDLE state (enables OFF). Otherwise it is ignored.

The result of this command is written to EEPROM.

IMPORTANT: The new Mobile Mode parameters take effect immediately. However, unless the SET switch is set to the 0 position, Mobile Mode parameters will reset based on the SET switch position whenever input power to the SPS-1 is removed and restored.

- **SWMODE,m**
Switch mode setting

m= 0= Momentary Mode false. Enable switch is a toggle switch. Power enable is based on switch position.

1= Momentary Mode true. Enable switch is a momentary pushbutton. Power enable is toggled On to Off or Off to On with each single press.

If Mobile Mode is enabled and Timeout Timer is active, a double press resets the Timeout Timer to its starting value.

- **CONFIG** <no parameters>

Returns the current configuration of the SPS1 in the following format:

/fftt:SETTINGS,SPS1,xx,uuuuu,ooooo,cccc,a,t,m,moff,mon,mto,cal,ofst,s,y,z:XX

Where:

- ff= The current DCN Address of this device
- tt= The address of the device that sent the CONFIG command. (00 if received in a broadcast command)
- SETTINGS= The message type of this response
- SPS1= The name of this device
- xx= The 2 digit hex address currently programmed in EEPROM for this device.
NOTE: This is not necessarily the current address of the device.
- uuuuu= Undervoltage trip level in millivolts
- ooooo= Overvoltage trip level in millivolts
- ccccc= Overcurrent trip level in milliamps
- a= Autoreset setting. 0= Manual Reset 1= Automatic Reset
- t= Autoreset time delay in seconds
- m= Mobile Mode Status 1=Mobile Mode Enabled, 0=Mobile Mode Disabled
- moff= Mobile Mode Off Threshold in mv
- mon= Mobile Mode On Threshold in mv
- mto= Mobile Mode Timeout time in minutes
- cal= Calibration scaling for current sensing readout & trip level (See Calibration section)
- ofst= Calibration offset for current sensing readout & trip level (See Calibration section)
- s= Switch mode. 0= Momentary mode is false. 1= Momentary mode is true.
- y= Physical DCN ADDR Switch setting on power up.
- z= Physical SET Switch setting on power up.

- **VERSION** <no parameters (Version 1.1 and later)

Returns the current firmware version in the following format:

/fftt:FWVER,SPS1,<ver>:XX

Where:

- ff= The current DCN Address of this device
- tt= The address of the device that sent the CONFIG command. (00 if received in a broadcast command)
- SETTINGS= The message type of this response
- SPS1= The name of this device
- <ver>= The current firmware version as a string (10 characters max)

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- **CALLO,cccc**
Calibration Low command
cccc= measured current in milliamps

IMPORTANT: See Calibration section for details on using this command.

- **CALHI,cccc**
Calibration High command
cccc= measured current in milliamps

IMPORTANT: See Calibration section for details on using this command.

- **CALSET,ss.sss,ooooo (V1.1 firmware and later)**
ss.sss= Current reading scale factor. This is always a positive number.
ooooo= Current reading offset in milliamps. This may be positive or negative.

Calibration Set Command. This sets the SPS-1 current reading calibration to the values sent in the command. The command is provided to reload the calibration data in the event that the SPS-1 internal EEPROM settings get corrupted. The scale and offset values are the same values reported in the SETTINGS response to the CONFIG command.

You can use the CONFIG command to get these values and record them for safe keeping. You can also contact A2Z Tech and provide your SPS-1 serial number and the factory calibration values will be sent to you.

- **LOGS <no parameters>**
Returns the fault history of the SPS1 in the following format:

/fftt:HISTORY,SPS1,,tot,uv,ov,oc:XX<cr>

Where:

- ff = The address of this device
- tt = The address of the device that sent the LOGS command (or 00 if State was sent in a broadcast message)
- HISTORY = Response message identifier – Historical Data
- SPS1 = The name of this device
- Tot= The total amount of time the output has been ON in tenths of a second.
NOTE: This value can be up to 4,294,967,296 tenths (~ 13.5 years)
This value is NOT reset by the RSTLOGS command.
- uv= The number of Undervoltage trips since production or since data was reset
- ov= The number of Overvoltage trips since production or since data was reset

- uc= The number of Overcurrent trips since production or since data was reset

SEE: RSTLOGS Command

- **SETADDR,xx**

Set the DCN Address of this device

xx= 1 or 2 hex digits

The SPS1 internally stores it's address as a single byte of data. Therefore it supports any one of 255 addresses from 01 through FF. It will not accept an address of 00.

The result of this command is stored to EEPROM. The current DCN address of this device is also **immediately set to the commanded address.**

IMPORTANT: In order for this address to persist after power is removed from the SPS1, the ADDR switch must be set to '0', otherwise the DCN Address will revert to the address set on the switch after power is removed and re-applied to the SPS1.

- **RSTLOGS** <no parameters>

This command resets the stored fault memory of the device to zero for each fault type.

NOTE: This does not reset the total accumulated ON time value.

- **REBOOT** <no parameters> **(Version 1.1 firmware and later)**

This command reboots the SPS-1 as if a power cycle had occurred.